

Speech Signals Complement LLMs for Predicting Interpersonal Attraction in Speed Dating

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If an LLM can read the whole transcript, does speech still add anything?

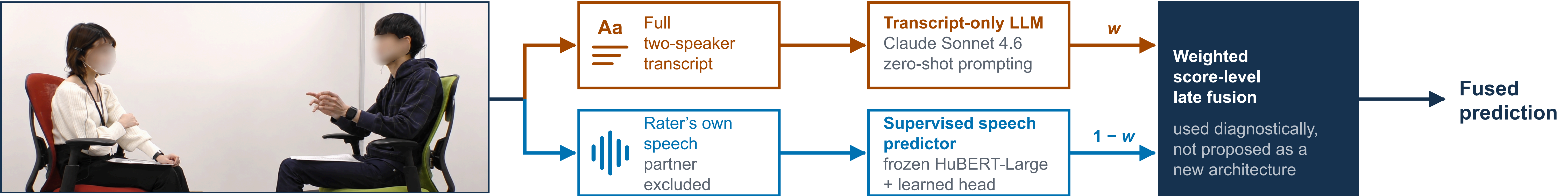
Yes — for ordering partners.

Speech-derived predictions still add value after the LLM has read the full transcript.

Fusion improves partner ranking over the transcript-only LLM in all four conditions ($\Delta PW = +.025$ to $+.054$; all Holm-corrected $p < .05$).

01 SETUP

The LLM reads both speakers' words. The supervised speech predictor uses only the rater's own speech.



One speed-dating conversation — source of both inputs

CORPUS

- 147 participants
- 1,248 conversations, Japanese face-to-face speed dating
- 1–9 13-item liking score, rated after each conversation

WHAT IS LEARNED?

- Claude** · no task-specific training; zero-shot prompting. Extended thinking is disabled.
- HuBERT branch** · HuBERT-Large is frozen. Additive attention pooling and a two-layer regression head are trained with CCC.
- Fusion** · nothing is trained end-to-end; only the scalar w is fitted.

CONDITION LABELS

- S1** Session 1 — first encounter, ≈ 5 min
- S2** Session 2 — same partner again, ≈ 10 min
- F2M** Female rates the male partner
- M2F** Male rates the female partner

SUPERVISED BASELINE CHECK

- Mean per-participant r (point estimates), all four conditions:
- Claude** > Sentence-T5, BERT, and J-LIWC
- Claude + HuBERT** > Sentence-T5 + HuBERT late fusion

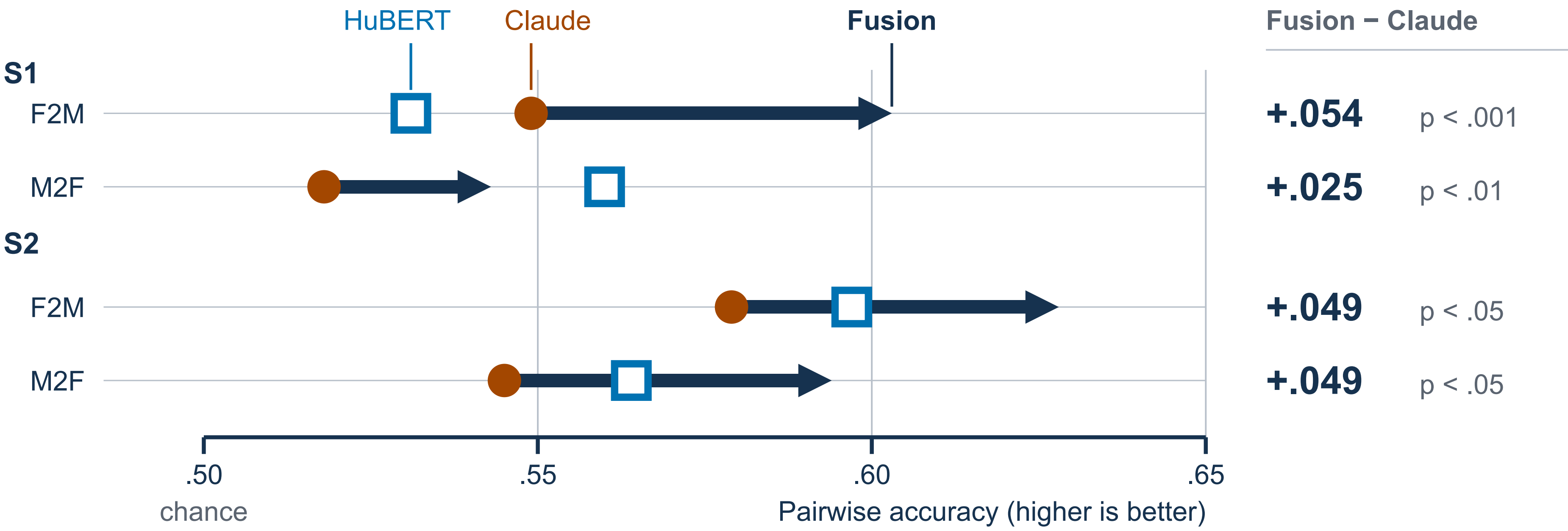
METRICS

- Both ask which partners a participant liked more.
- Pairwise accuracy (PW)** asks pair by pair — order only.
- Per-participant Pearson r** considers the whole pattern across partners — not only pairwise order.
- Fused score** = $w \times \text{Claude score} + (1 - w) \times \text{HuBERT score}$
- 25-fold leave-one-group-out. The fusion weight w is one scalar per condition and test fold, selected by CCC on pooled out-of-fold predictions from the remaining folds — the test fold is never used. The paper's primary metric is per-participant r ; PW isolates ordering, and that is where the gain is robust. Full method: QR.

02 MAIN RESULT

Fusion orders partners better than Claude in every round and rating direction.

Pairwise accuracy (PW): significant gain over Claude in all four conditions.



Arrow length is the gain of fusion over Claude; all p -values are Holm-corrected across the four conditions. S1 M2F: HuBERT exceeds fusion — the four-of-four claim compares fusion with Claude. Rank-biserial r in plotted order: .594, .623, .299, .374.

Per-participant r : no uniform gain.

Mean per-participant r	S1 F2M	S1 M2F	S2 F2M	S2 M2F
Claude	.248	.163	.244	.238
HuBERT	.105	.092	.212	.183
Fusion	.281	.164	.323	.248
Fusion – Claude	+.033	+.001	+.079	+.010

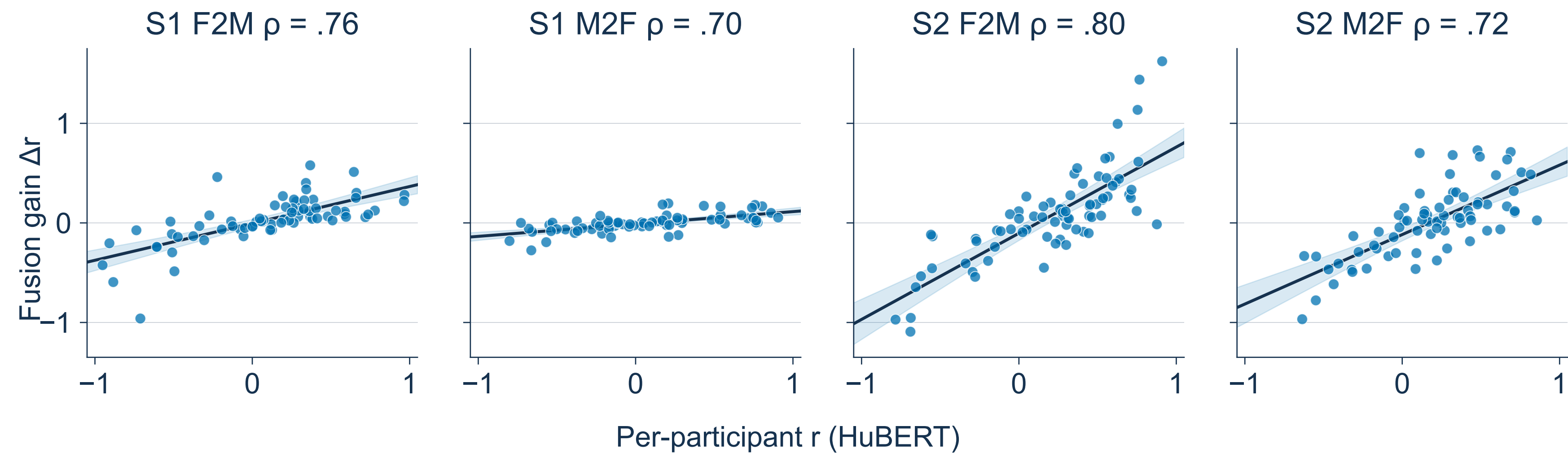
No paired Δr survives Holm correction.

Claude and HuBERT share little within-rater variance.

HuBERT beyond Claude	Claude beyond HuBERT
+.14 – +.21	+.15 – +.24
Shared R^2	Within-rater, in-sample incremental R^2 ; distinct information and noise reduction are not separated.
.01 – .04	

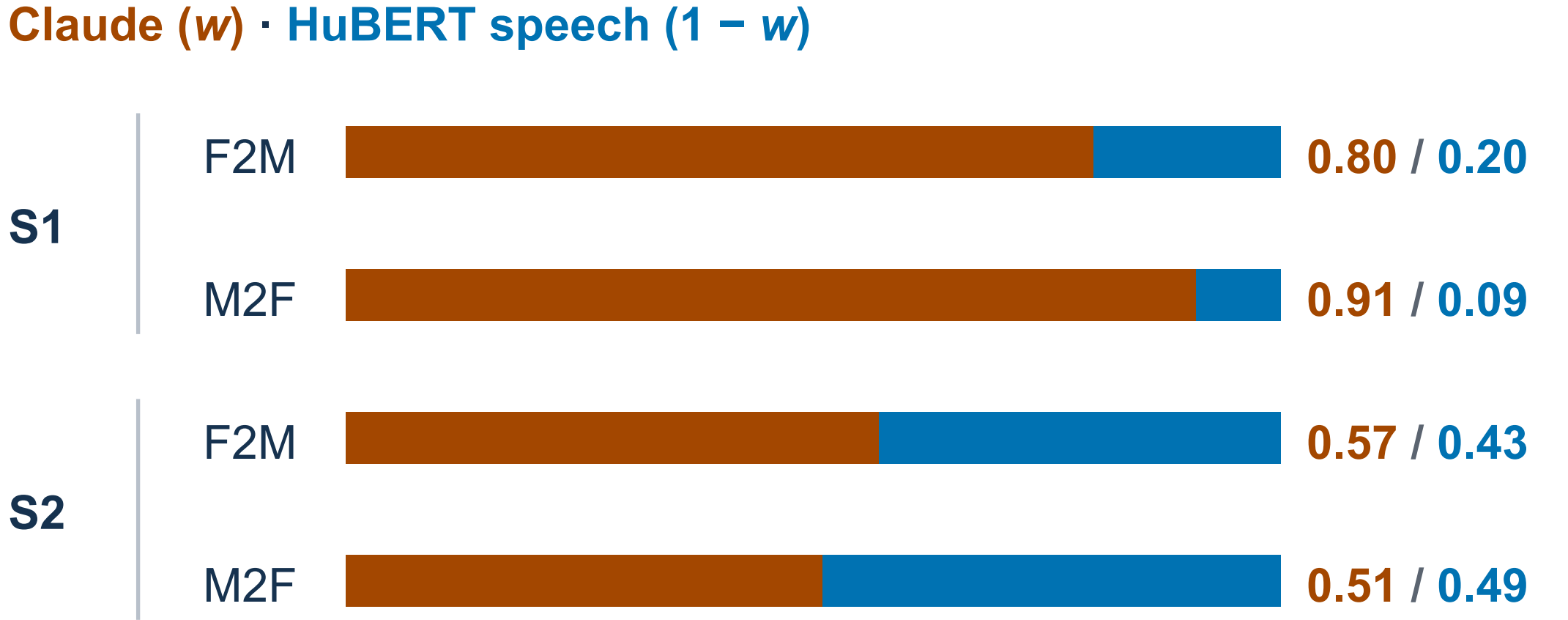
03 SUPPORTING ANALYSES

Fusion Δr is larger for participants with higher HuBERT per-participant r .



Points are participants; lines and bands are OLS fits with 95% CIs. $\rho = .70$ –.80 in all four conditions, all $p < .0001$. Retrospective and partly structural, since HuBERT enters the fused score.

Fusion is Claude-heavy in S1 and nearly balanced in S2.



Mean selected weights across 25 folds, selected per condition and fold — **not per participant**. Fold-selected weighting adjusts each branch's contribution instead of forcing an equal average.

04 TAKE-AWAY

- Reading the whole transcript does not make the rater's voice redundant: fusion improves partner ordering over the LLM in all four conditions.
- The gain is in ordering; it does not extend uniformly to per-participant r , and the acoustic or interactional cues remain unidentified.
- Next: identify those cues; estimate speech-branch reliability prospectively.

Scope: one Japanese opposite-sex speed-dating corpus; absolute performance is modest and not deployment-ready.



Paper